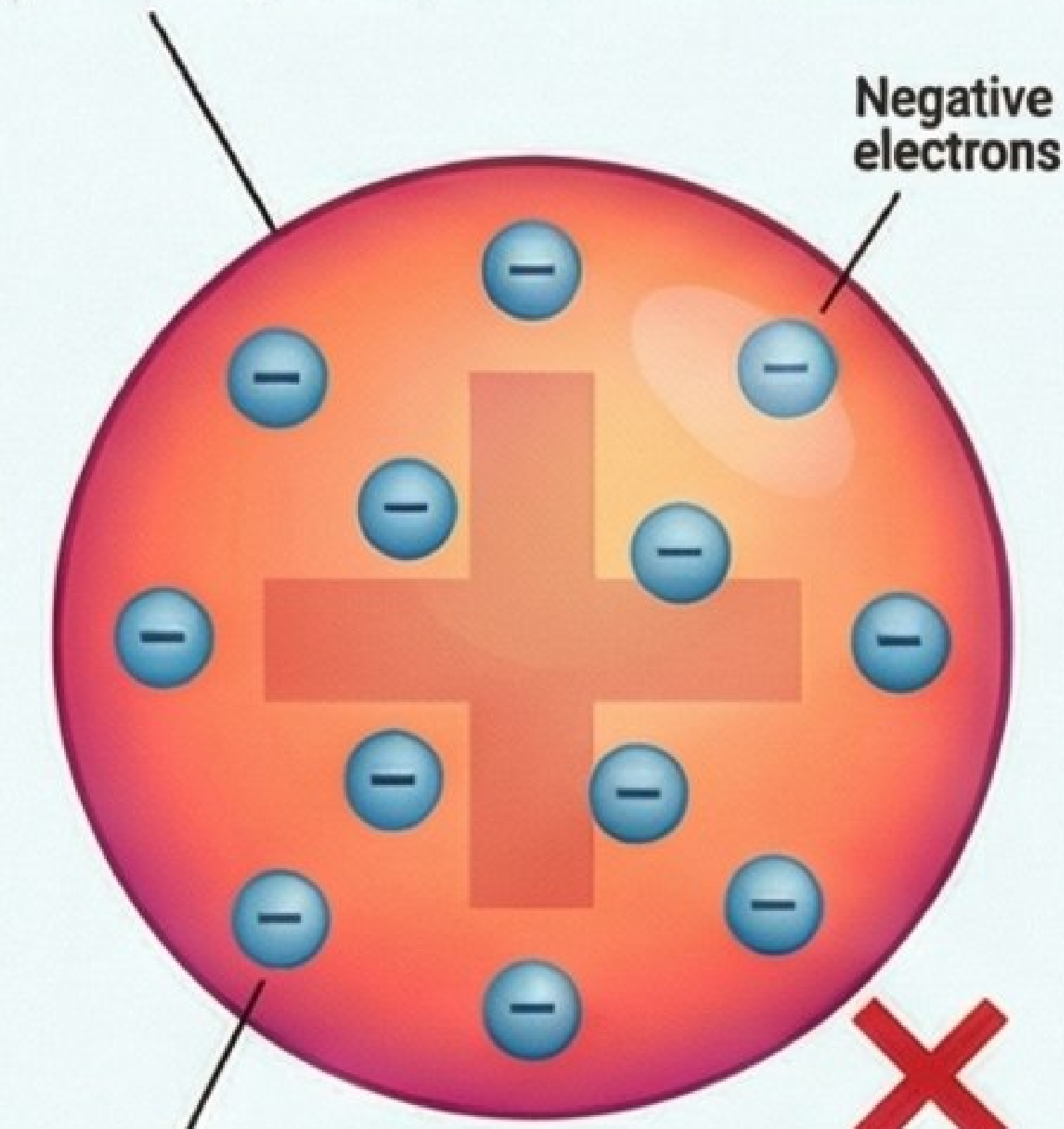


The Evolution of the Atom: From Plum Pudding to Bohr's Orbits

1. The Early Idea: Thomson's Model

The "Plum Pudding" Model:
Proposed that an atom is a uniform sphere of positive charge with negatively charged electrons embedded throughout, similar to plums in a pudding .. or seeds in a watermelon.

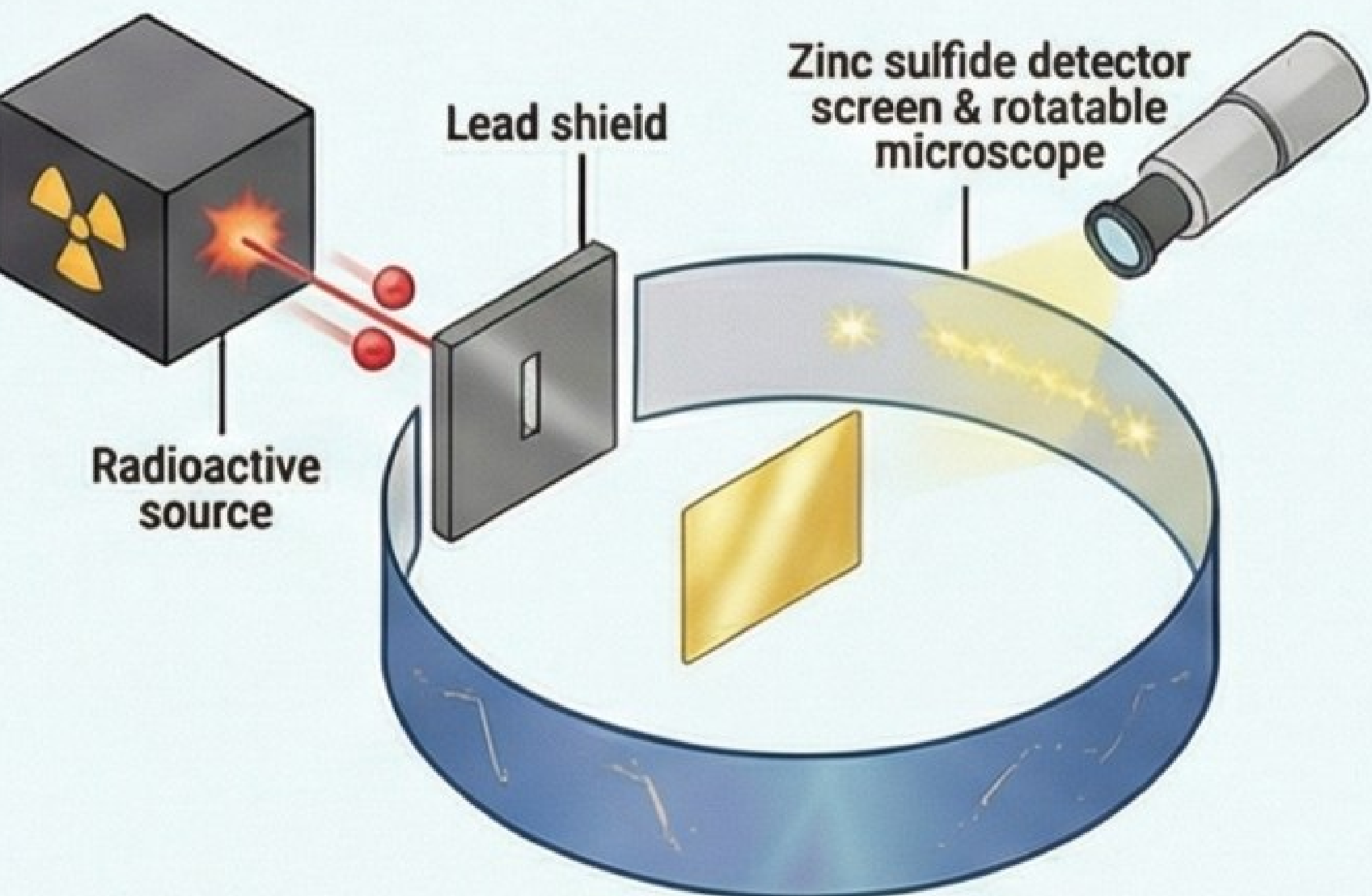


Electrically Neutral:
The total positive charge of the sphere was balanced by the total negative charge of the electrons, resulting in a neutral atom.

Why It Failed:
This model could not explain the results observed in Rutherford's subsequent gold foil experiment.

2. The Game Changer: Rutherford's Gold Foil Experiment

The Experimental Setup:
Alpha particles from a radioactive source were fired at a very thin gold foil, with a zinc sulfide screen and a rotatable microscope to detect where the particles landed.



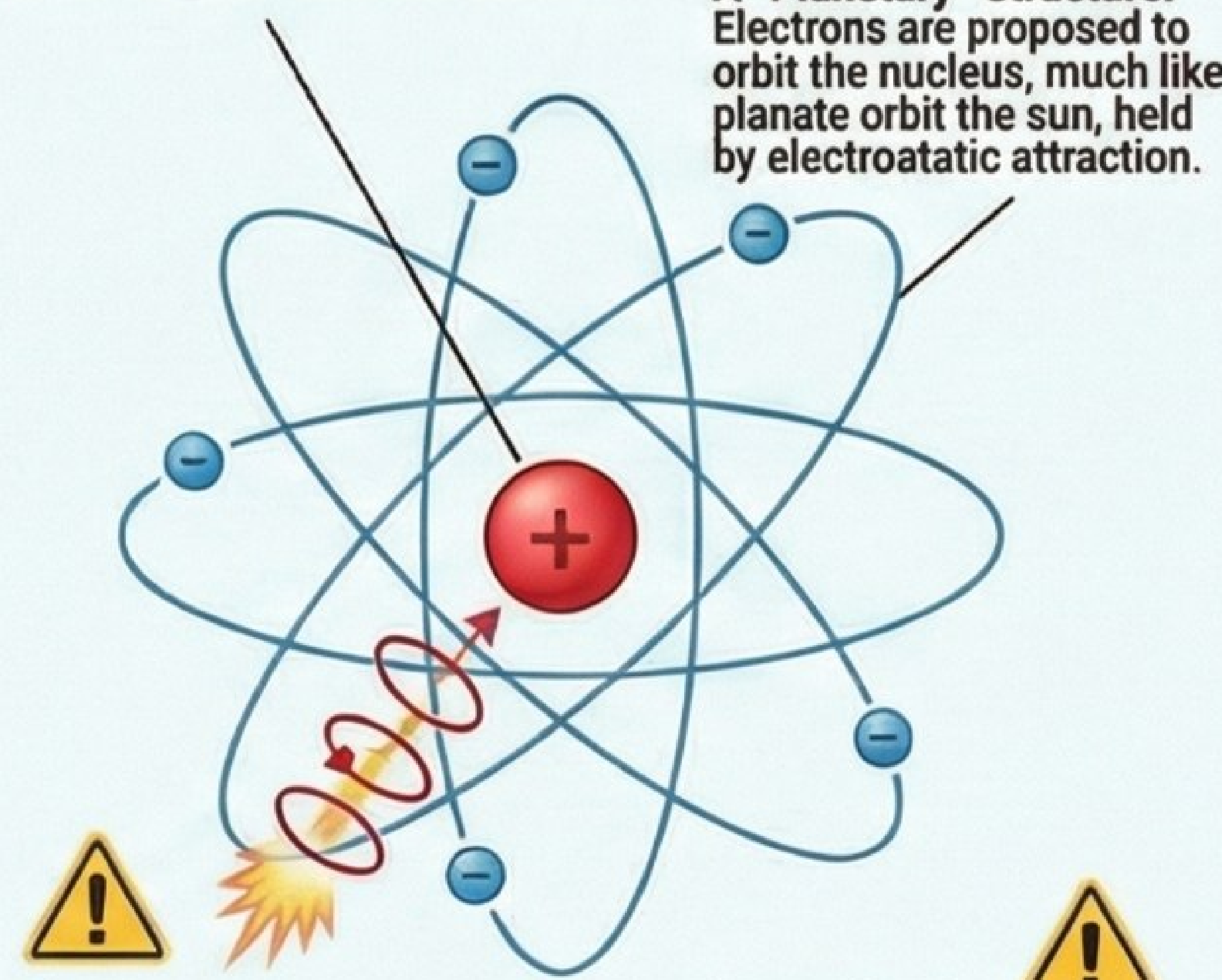
Observation 1: Most Particles Passed Straight Through: The vast majority of alpha particles went through the foil with little to no deflection ($\pm 1^\circ$), suggesting the atom is mostly empty space.

Observation 2: Some Particles Were Deflected: About 1 in 8000 particles were deflected by significant angles, indicating they encountered a repulsive force.

Observation 3: A Tiny Fraction Bounced Back: An extremely small number of particles (0.14% scattered $< 30^\circ$, with a mere few at 180°) were deflected back, suggesting a collision with something very dense and positively charged.

3. The Nuclear Model: Rutherford's Atom & Its Limitations

The Atom has a Dense, Positive Core:
Rutherford concluded that all the positive charge and most of the atom's mass are concentrated in a tiny central region called the nucleus.



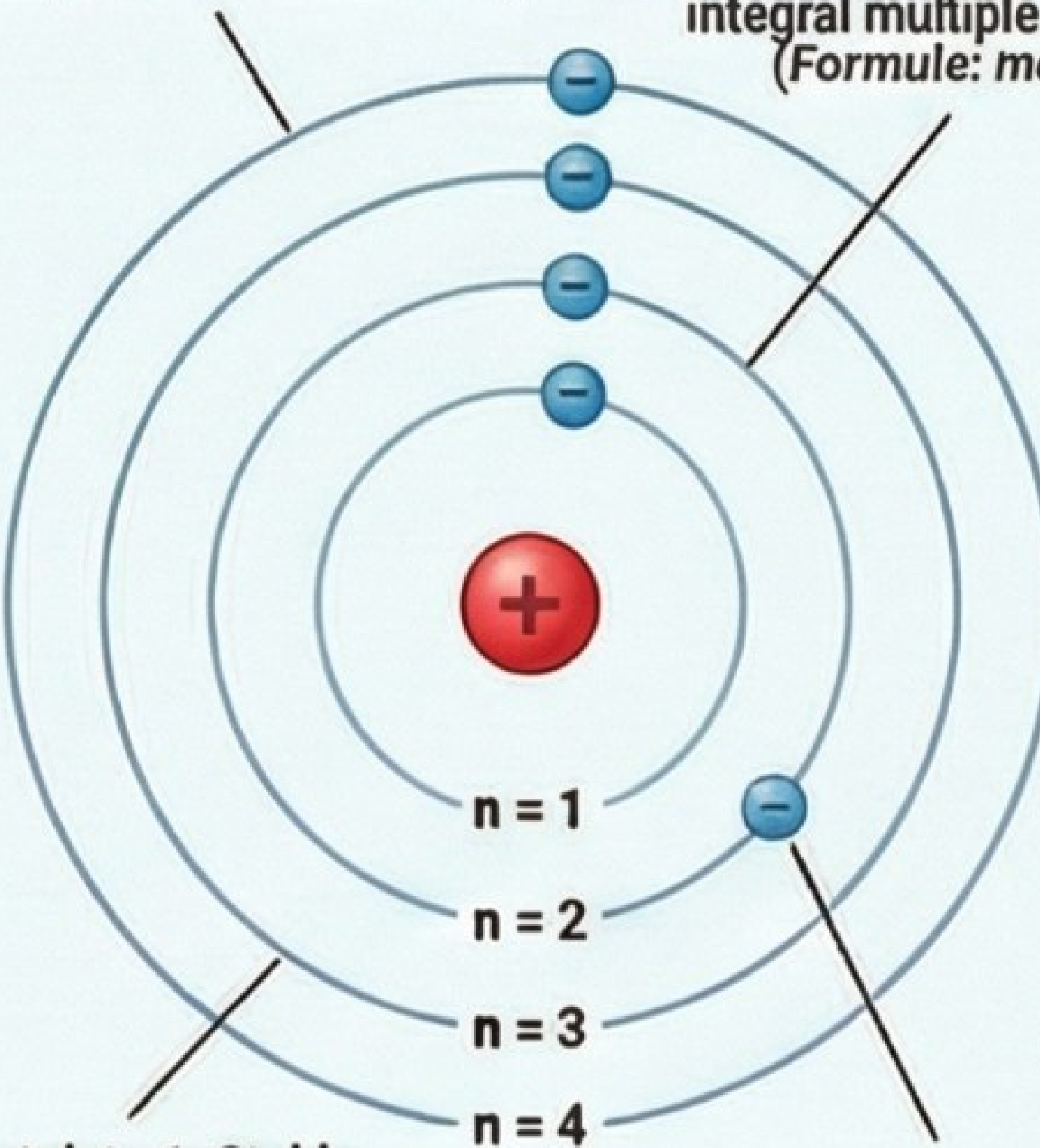
A "Planetary" Structure:
Electrons are proposed to orbit the nucleus, much like planets orbit the sun, held by electrostatic attraction.

Flaw 1: The Instability Problem:
According to classical physics, an orbiting electron should continuously radiate energy, lose speed, and spiral into the nucleus, causing the atom to collapse.

Flaw 2: The Line Spectroscopy Mystery:
The model could not explain why atoms emit light only at specific, discrete wavelengths (line spectra) when heated.

4. The Quantum Leap: Bohr's Atomic Model

Postulate 1: Stable Orbits:
Electrons revolve in specific, stable orbits called "stationary orbits" where they do not radiate energy.



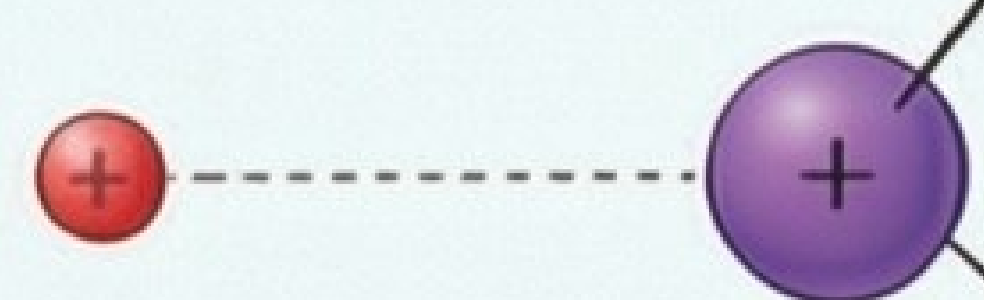
Postulate 3: Quantized Angular Momentum:
An electron can only be in an orbit where its angular momentum (mvr) is an integral multiple of $\hbar/2\pi$.
(Formula: $mvr = n\hbar$)

Postulate 1: Stable Orbits:
Electrons revolve in specific, stable orbits where they do not radiate energy.

Postulate 3: Energy Transitions:
An electron emits or absorbs energy (as a photon) only when it jumps between these allowed orbits. The photon's energy equals the energy difference between the orbits ($E = E_2 - E_1$).

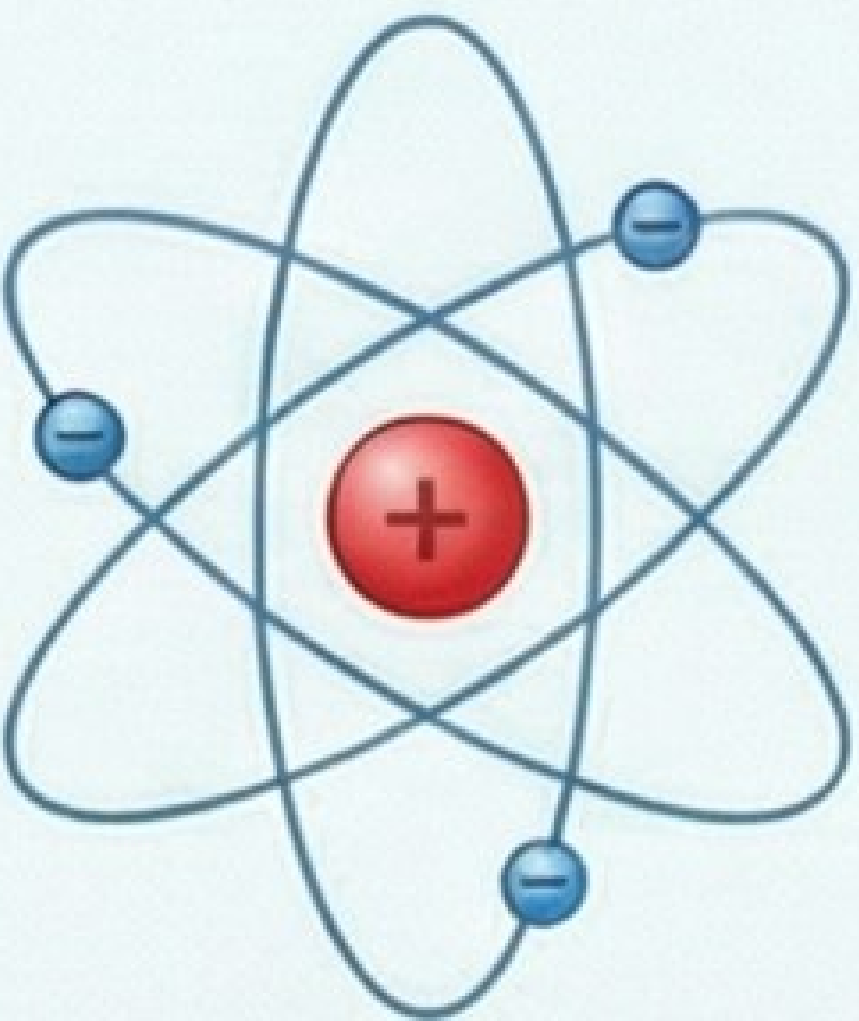
5. Understanding Atomic Behavior: Key Concepts & Spectra

Key Concepts

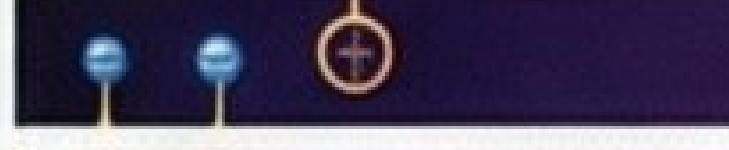
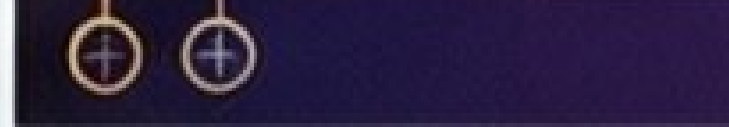


Distance of Closest Approach: The minimum distance an alpha particle gets to the nucleus before its kinetic energy is fully converted to potential energy and it is repelled back. This helps estimate the size of the nucleus.

Impact Parameter: The perpendicular distance between the path of an alpha particle and the center of the nucleus. A smaller impact parameter results in a larger scattering angle.



Atomic Spectra: When atoms are excited, electrons jump to higher orbits and then fall back, emitting light of specific wavelengths. This creates a unique "line spectrum" for each element.

	Series Name	Final Orbit (n_1)	Originating Orbits (n_2)	Spectral Region
Lyman		1 (Innermost)	2, 3, 4... ∞	Ultraviolet
Balmer		2	3, 4, 5... ∞	Visible
Paschen		3	4, 5, 6... ∞	Infrared
Brackett		4	5, 6, 7... ∞	Infrared
Pfund		5 (Outermost)	6, 7, 8... ∞	Far Infrared